
Smarter Medicine: How AI is Transforming Radiation Treatment

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1 Introduction: A Glimpse into Radiation Medicine

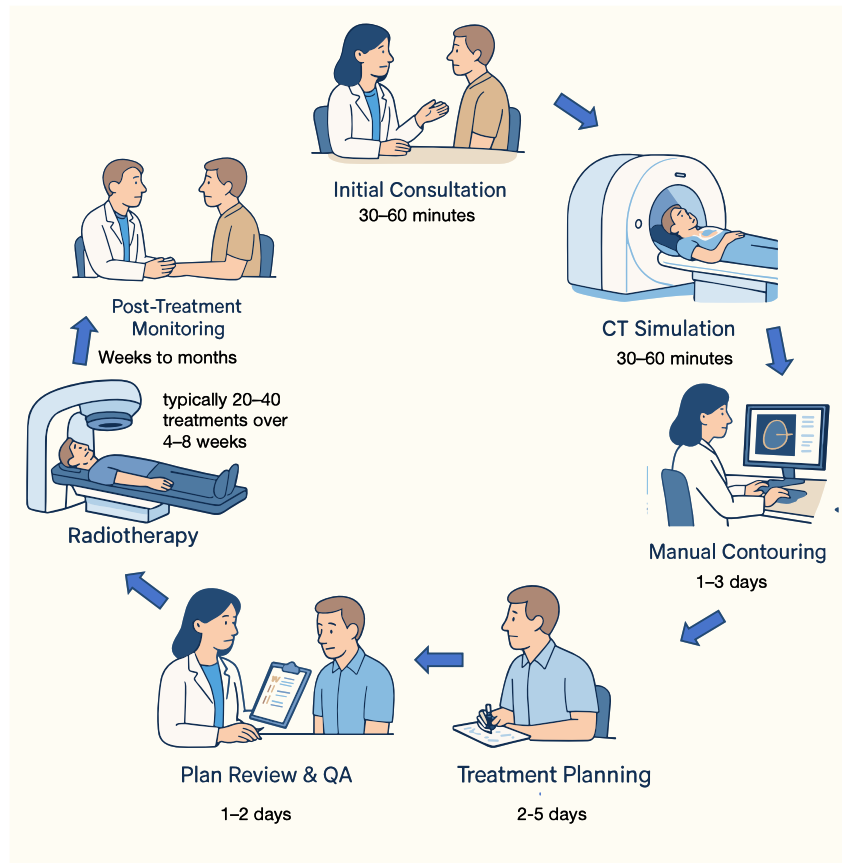


Figure 1: Current Radiation Treatment Workflow

While the fundamentals of radiation oncology are covered extensively in other chapters of this book, in this chapter we shift our focus to how artificial intelligence (AI) is reshaping its clinical workflow. From medical image processing to treatment planning, AI technologies are improving workflow efficiency and enabling more precise, individualized cancer treatment.

6 To illustrate the typical patient experience and highlight where AI could make a meaningful difference,
7 let us follow Mr. Harrison's journey as he navigates the complex process of receiving radiation
8 therapy for early-stage prostate cancer.

9 Mr. Harrison's care journey began with an initial consultation with radiation oncologist Dr. Anya
10 Sharma. She outlined the treatment options for early-stage prostate cancer, focusing on External
11 Beam Radiation Therapy (EBRT), and explained potential side effects. Although the conversation
12 was complex, it reflected the highly personalized and multidisciplinary nature of modern radiation
13 therapy.

14 The next critical step was CT simulation, which forms the anatomical foundation for treatment
15 planning. Mr. Harrison was positioned using a custom-molded immobilization device to ensure
16 reproducibility across treatment sessions. Technicians placed small fiducial markers on his skin to
17 guide future beam alignment. As the CT scanner rotated around him, high-resolution imaging data
18 were acquired to map his internal anatomy and define target structures. The accuracy of this step
19 is crucial—errors in imaging or positioning can lead to suboptimal dose delivery or unnecessary
20 exposure to nearby Organs at Risk (OARs).

21 The subsequent treatment planning phase was resource-intensive and requires close collaboration
22 among radiation oncologists, dosimetrists, and physicists. Clinicians manually delineated the tumor
23 and surrounding OARs, while the planning team iteratively adjusted beam geometries and dose
24 distributions to create an optimal radiation plan. This plan then underwent rigorous quality assurance
25 before treatment began. Due to the complexity of these steps, Mr. Harrison had to wait several weeks
26 between simulation and the start of treatment.

27 Over the course of several weeks, Mr. Harrison underwent daily radiation sessions. Each day, he was
28 positioned using image guidance, followed by precise dose delivery. After completing treatment, he
29 returned for regular follow-up visits to assess therapeutic response and manage any late side effects.
30 While effective, the overall process demanded substantial time and expert input, underscoring the
31 need for solutions that can improve efficiency, consistency, and quality of care.

32 Artificial intelligence is rapidly emerging as a solution to these challenges. By leveraging large-scale
33 medical datasets and expert-annotated labels, deep learning models enable automation and decision
34 support throughout the treatment pipeline. For instance, AI-driven auto-contouring algorithms can
35 rapidly and reproducibly delineate tumors and OARs, reducing both inter-observer variability and
36 manual workload. Intelligent planning systems generate high-quality treatment plans with minimal
37 human input. Furthermore, AI-powered platforms support adaptive radiotherapy by modifying
38 treatment plans in response to daily anatomical changes. In short, these innovations are reshaping
39 radiation oncology into a more precise, standardized, and efficient model of care.

40 **2 What is Artificial Intelligence?**

41 Artificial intelligence (AI) is becoming increasingly integrated into daily life. For example, tools
42 like ChatGPT can generate human-like text, answer complex questions, and assist with writing tasks.
43 Meanwhile, self-driving cars developed by companies such as Tesla and Waymo can perceive their
44 surroundings and make real-time driving decisions without human intervention. These advances
45 demonstrate that AI systems are now capable of performing functions that once required human
46 intelligence, including language understanding, decision-making, and problem-solving. Although
47 these applications represent only a fraction of AI's potential, they reflect a broader trend: AI is
48 beginning to reshape many aspects of modern society.

49 How should AI be defined? To understand it, let us go back to 1950. In his work *Computing*
50 *Machinery and Intelligence*, Alan Turing proposed the famous question "Can machines think?". In
51 Turing's perspective, a machine can think if it may fool a human spectator that it is a human through
52 direct written conversation. This is also known as the Turing Test. Though language mimicry alone
53 no longer defines AI today, the core idea remains: intelligence involves learning, reasoning, and
54 adapting based on input data.

55 How do we build an AI model? Most modern AI systems are developed through a process known
56 as training, where algorithms learn to identify patterns and make predictions based on data. This
57 learning paradigm is loosely inspired by how humans acquire knowledge from experience. At the core
58 of many AI models are neural networks, which consist of layers of interconnected parameters. During

training, the model receives input data and produces an output, such as a classification or prediction, which is then compared against a known target. The error between the predicted and actual outputs is quantified using a loss function. This loss is then used to adjust the model's internal parameters through an optimization algorithm, most commonly based on gradient descent. By repeating this process over many examples, the model gradually improves its accuracy and learns a functional mapping from inputs to outputs.

Example: Convolutional Neural Networks (CNN)

One of the most influential AI architectures is the Convolutional Neural Network (CNN), which was introduced in the 1980s and is still widely used today.

To understand CNNs, it is helpful to begin with a basic building block of neural networks: the neuron. In a typical feedforward network, such as a Multilayer Perceptron (MLP), each neuron applies a simple linear transformation to its input:

$$Y = \beta_1 X + \beta_2$$

where β_1 and β_2 are parameters learned during training. In deep networks, many such neurons are stacked into layers, and the number of parameters grows quickly. This can make learning inefficient and difficult to generalize, especially for high-dimensional inputs like images.

CNNs address this by introducing the concept of local receptive fields, inspired by how the human visual system works. In our eyes, individual neurons respond only to a small region of the visual field (which is called their receptive field), and the brain integrates these local signals to form a complete image. Similarly, a CNN uses small filters (called kernels) that scan across the image, focusing on local patterns such as edges or textures. These kernels are applied repeatedly across the entire image, and their parameters are shared, which greatly reduces model complexity while preserving the ability to learn spatial features.

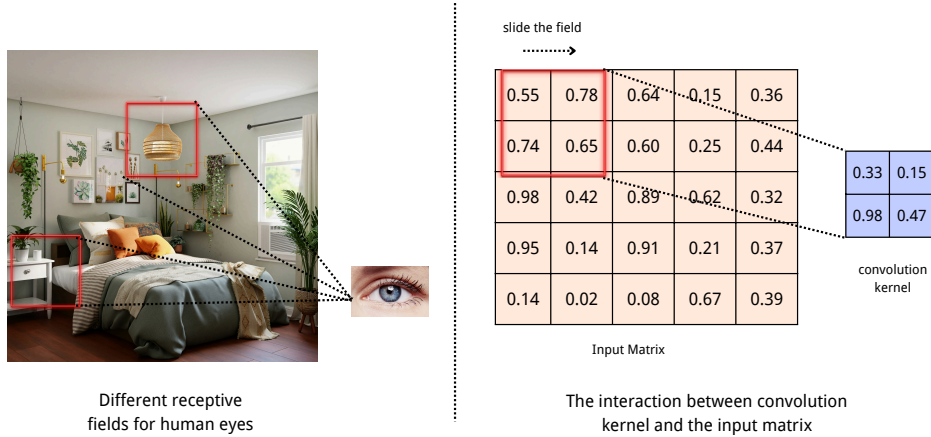


Figure 2: Analogy between human receptive fields and CNN convolutional kernels

It is important to note that the example above is highly simplified. In practice, a complete AI model involves many intermediate steps, large numbers of parameters, and complex mathematical operations. Researchers must carefully decide on several key components: how to design the mnetwork architecture, which loss function to use, how to adjust the parameters at each iteration, and how many iterations are needed. These design work together to determine the model's overall performance.

In radiation medicine, AI holds significant promise across multiple stages of the clinical workflow. For example, medical images such as CT and MRI scans are routinely used to localize tumors, assess

89 treatment response, and guide therapy planning. AI algorithms, trained on these image datasets,
90 can assist in these tasks. These capabilities not only reduce clinician workload but also improve
91 consistency and precision in care delivery. While challenges such as reliability and clinical validation
92 remain, ongoing research continues to push the boundaries of what AI can achieve in oncology. As
93 evidence accumulates, AI is increasingly positioned as a valuable tool in enhancing the quality and
94 efficiency of radiation therapy.

95 **3 Medical Image Segmentation**

96 **3.1 Clinical Scenario Definition**

97 Medical image segmentation refers to the process of identifying and delineating specific anatomical
98 structures (such as organs, tumors, or tissues) within CT, MRI, or other imaging modalities. In
99 essence, it involves tracing the exact boundaries of these structures to separate them from surrounding
100 anatomy.

101 Why is this level of precision so critical? Consider the case of a lung cancer patient receiving radiation
102 therapy. The tumor must be completely eliminated to ensure treatment success. However, nearby
103 vital organs such as the heart, spinal cord, and healthy lung tissue must be carefully spared. Even a
104 slight extension of radiation beyond the tumor boundary can damage these critical structures and lead
105 to severe complications.

106 This creates a medical paradox: you must be aggressive enough to eliminate the cancer, yet gentle
107 enough to preserve the delicate structures that keep your patient alive and functioning. Thus, without
108 precise boundaries, you risk trading one life-threatening condition for another.

109 This paradox underscores the importance of precise medical image segmentation in oncology. By
110 accurately mapping both the target areas (regions of interest or ROIs) and the vulnerable structures
111 (organs at risk or OARs), clinicians can create a detailed treatment plan. The computer uses these
112 outlined boundaries to aim radiation beams exactly at the tumor while avoiding nearby healthy organs.
113 Instead of one large beam that would damage everything in its path, multiple smaller beams from
114 different angles meet precisely at the tumor location, delivering a powerful dose to the cancer while
115 sparing surrounding tissues.

116 Despite its clinical value, manual segmentation remains a major bottleneck. Experts must often
117 annotate hundreds of 2D slices to cover a full 3D scan, which can take several hours per case. The
118 process is not only time-consuming, but also subject to inter-observer variability. Moreover, the
119 repetitive and detail-oriented nature of the task makes it prone to fatigue and human error. These
120 limitations highlight the value of AI-driven segmentation in clinical workflows.

121 **3.2 AI-Driven Segmentation**

122 As introduced earlier, AI models learn by training on large amounts of labeled data. In the context of
123 medical image segmentation, this means providing the model with CT, MRI, or other medical images
124 as inputs, along with expert-annotated segmentation masks as labels. These masks define the exact
125 boundaries of structures such as organs or tumors, enabling the model to learn how to automatically
126 delineate them in new, unseen scans.

127 One of the most widely used AI models for medical image segmentation is **U-Net**, a neural network
128 architecture specifically designed for this task. Its name comes from its U-shaped structure, which
129 reflects how the network processes images in two stages: first by compressing the image to understand
130 the overall context, then by expanding it to recover the fine details needed for precise segmentation.

131 The first part of U-Net, known as the *contracting path*, works like zooming out to get the “big picture.”
132 The network gradually reduces the image resolution while learning important features such as edges,
133 textures, or patterns that help it understand what is present in the image.

134 The second part, called the *expanding path*, functions like zooming in. In this stage, the network
135 restores the image to its original resolution. Rather than generating a simple copy, it produces a
136 detailed segmentation map that classifies each pixel into categories such as tumor, liver, lung, or
137 background.

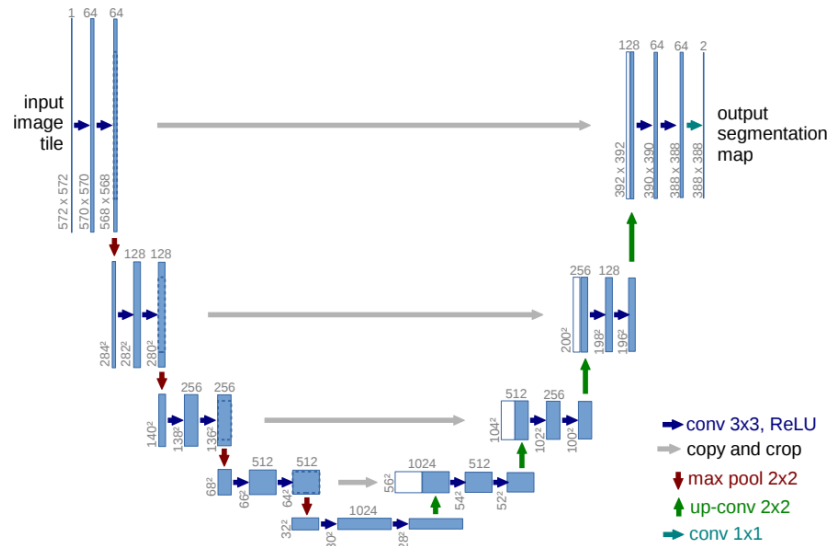


Figure 3: Architecture of U-Net, consisting of a contracting path (left) and an expansive path (right). Figure adapted from Ronneberger et al. [1].

What makes U-Net especially powerful is its use of **skip connections**—shortcuts that link the zoom-out and zoom-in parts of the network. These connections allow the model to reuse fine details that might otherwise be lost during the compression stage. In medical imaging, the difference between healthy and diseased tissue can be extremely subtle, so retaining these details is essential.

Once trained, AI-based segmentation model can rapidly process new medical images and generate accurate segmentation results within seconds. By automating what was traditionally a time-consuming and labor-intensive task, AI-based segmentation significantly accelerates clinical workflows. This not only shortens the time required for treatment planning but also reduces the manual workload on clinicians, enabling them to focus more on diagnosis, decision-making, and patient care.

4 Dose Prediction

4.1 Clinical Scenario Definition

In radiation therapy, after the tumor and nearby organs at risk (OARs) have been carefully delineated, clinicians must determine how radiation should be delivered to treat the cancer effectively while avoiding harm to healthy tissues. This step, known as dose prediction or dose planning, involves estimating the spatial distribution of radiation dose throughout the patient's body.

The goal is to deliver a high dose to the tumor while minimizing exposure to nearby sensitive structures. Achieving this balance is challenging: the dose must be sculpted precisely around complex anatomy, often requiring time-consuming manual adjustments and extensive clinical expertise.

This process becomes even more complex with modern techniques such as intensity-modulated radiation therapy (IMRT), where radiation is delivered from multiple angles with varying intensities. Generating a high-quality dose plan can take considerable time and varies depending on the planner's experience.

Given these challenges, artificial intelligence has emerged as a promising tool to support dose planning. By learning from large datasets of prior treatment plans and patient anatomy, AI models can capture patterns in dose distribution and generate clinically acceptable predictions within seconds. These tools have the potential to accelerate the planning process, reduce inter-physician variability, and support more standardized and personalized treatment delivery.

4.2 AI-Driven Dose Prediction

AI-driven dose prediction aims to automatically estimate a clinically acceptable 3D radiation dose distribution based on a patient’s medical scans and delineated structures. The input to such models typically includes volumetric CT/MRI images and binary masks representing regions of interest, such as the planning target volume (PTV) and surrounding OARs. The output is a 3D dose distribution matrix aligned with the same spatial coordinates. These predictions can serve as references for clinicians, helping them evaluate possible dose plans more efficiently. The overall pipeline of AI-driven dose prediction is demonstrated in Figure 4.

Deep learning has played a central role in dose prediction, with a range of architectures developed to capture the complex relationship between anatomy and radiation dose distribution. Early work such as DoseNet [2] employed 3D CNNs to directly estimate dose distributions from anatomical input. More recently, transformer-based methods[3] have been explored to further improve long-range dependency modeling. In parallel, generative models such as DoseGANs[4] and DoseDiff[5] have also begun to emerge in this domain, aiming to enhance realism and uncertainty modeling in predicted dose maps.

Once trained, these models can be integrated into the clinical workflow. Physicians provide the patient’s scan and structural contours, and the AI system generates a proposed dose distribution. This prediction acts as a starting point that clinicians can review and adjust, potentially reducing planning time and improving inter-physician consistency. As more data becomes available, these models can continue to learn and adapt, enhancing performance over time.

Despite these advantages, AI-based dose prediction methods still faces several challenges. First, the accuracy and generalizability of models depend heavily on the diversity and quality of training data. Biases in the dataset, underrepresentation of certain patient groups, or variations in contouring practices across institutions may impact reliability. Furthermore, current AI models do not explicitly enforce physical constraints such as dose falloff patterns, machine delivery limitations, or energy attenuation laws, which may lead to clinically infeasible results without expert oversight. This limitation is partly due to the “black-box” nature of many deep learning models, which makes it difficult to interpret or control their internal decision-making processes. In addition, broader concerns including data privacy, ethical responsibility, and integration into clinical workflows must be carefully addressed to ensure safe and responsible deployment.

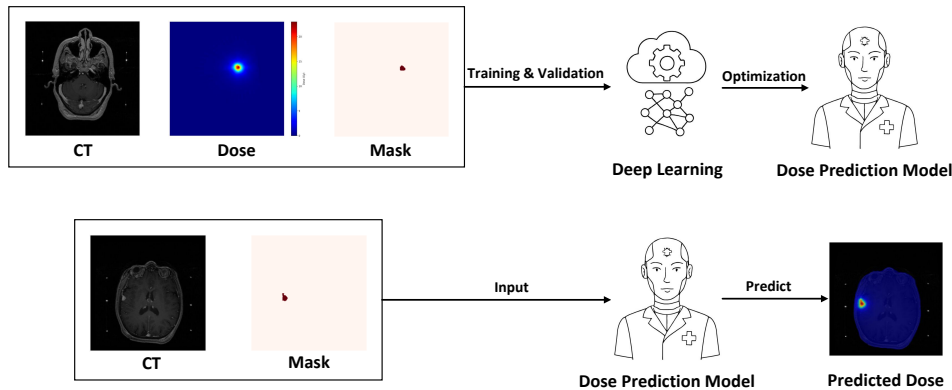


Figure 4: The overall pipeline of AI-driven dose prediction model for brain tumor radiation. We first train the model using annotated datasets (including CT, masks, and dose distribution) and apply the best-performance model on unseen cases to predict the dose.

5 Data Generation

5.1 Data Scarcity, Ethics

As discussed earlier, AI models rely fundamentally on data. Just as students learn more effectively with high-quality textbooks, AI systems require sufficient and well-curated data to produce meaningful

199 results. Without sufficient data, even the most advanced algorithms risk poor accuracy, overfitting,
 200 and reduced clinical utility.

201 However, in healthcare, data scarcity remains a persistent challenge. Medical data are tightly regulated
 202 due to privacy concerns and are expensive to acquire. Imaging procedures such as MRI scans incur
 203 high costs, and expert annotations require time-intensive input from trained clinicians. Additionally,
 204 most datasets exhibit *class imbalance*: common diseases and anatomies are overrepresented, while
 205 rare but clinically significant conditions are often lacking. This imbalance limits the model’s ability
 206 to generalize and perform reliably across diverse patient populations.

207 To mitigate these issues, synthetic data has emerged as a promising strategy. While synthetic data
 208 may seem artificial, when generated through rigorous methods, it can replicate the statistical and
 209 anatomical properties of real-world cases[6]. When properly validated, such data can be used to
 210 augment limited datasets, improve model robustness, and support rare condition modeling, addressing
 211 some of the core limitations of traditional data collection.

212 A widely adopted approach for generating such synthetic data is the **Generative Adversarial**
 213 **Network (GAN)**[7]. A GAN consists of two neural networks:

- 214 • The Generator, which produces synthetic images, such as artificial MRI or CT scans;
- 215 • The Discriminator, which attempts to distinguish real medical images from the generated
 216 ones.

217 These two networks engage in a dynamic adversarial process. The generator continuously improves
 218 by learning to fool the discriminator, while the discriminator becomes better at spotting fakes. Over
 219 time, the generator produces synthetic images that are increasingly realistic and clinically plausible.

220 This adversarial training allows GANs to generate diverse and high-fidelity medical data, simulating
 221 rare or underrepresented conditions, and helping to balance class distributions. Importantly, because
 222 the data is artificially created, it avoids the privacy risks associated with real patient information.
 223 When validated properly, synthetic data can thus serve as a trustworthy and effective complement to
 224 real-world datasets.

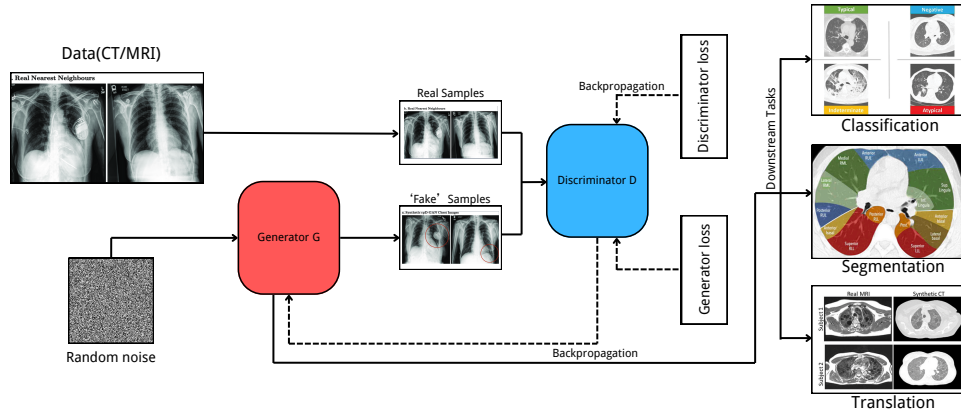


Figure 5: Pipeline for GAN-based medical image generation and its application to downstream tasks.

225 5.2 CT / MRI Conversion Synthesis

226 One practical application of synthetic data generation is cross-modality image synthesis[8], where AI
 227 models are trained to convert between different imaging modalities. This technique not only augments
 228 limited datasets but also enhances diagnostic utility by bridging the strengths of each modality.

229 In radiation medicine, two commonly used data types are CT and MRI images. CT is faster and
230 cheaper but less informative for soft tissues. MRI provides superior soft tissue contrast but is costly
231 and less accessible. If we could use AI to ‘translate’ CT images into synthetic MRI-like scans, we
232 would get the best of both worlds.

233 One particularly promising method is through **Diffusion Models**[9]. Diffusion models have recently
234 gained attention for their ability to generate highly realistic images. Unlike GANs, which generate
235 images in a single step through adversarial learning, diffusion models generate images through a
236 gradual denoising process.

237 The idea is intuitive:

- 238 • First, a real image is progressively corrupted by random noise.
- 239 • Then, the model is trained to reverse this process—to learn how to turn noise back into the
240 original image.

241 Once trained, the model can generate entirely new images by iteratively denoising pure noise. This
242 gradual, multi-step synthesis process produces stable outputs with fewer artifacts, making diffusion
243 models particularly well-suited for medical imaging tasks where anatomical accuracy and structural
244 fidelity are essential.

245 **5.3 Synthetic Data Is a Tool—Not a Replacement**

246 Synthetic data offers effective solutions to key challenges in applying AI to the medical domain,
247 including privacy concerns, data scarcity, and underrepresented clinical scenarios. However, its
248 limitations must be acknowledged. Synthetic images are not real medical scans and must not be used
249 directly for clinical decisions.

250 At present, synthetic data is primarily used for research and model development. Its clinical adoption
251 demands rigorous validation and regulatory oversight. Nevertheless, when developed responsibly,
252 synthetic data holds promise as a foundational tool for enabling safer, more robust, and more inclusive
253 AI in healthcare.

254 **6 Conclusion & Future Outlook**

255 **6.1 Summary**

256 This chapter examined how artificial intelligence is reshaping clinical workflows in radiation medicine.
257 We began by outlining key challenges in current practice, including labor-intensive manual segmen-
258 tation, prolonged treatment planning, and variability in clinical decision-making. To address these
259 issues, we introduced AI-driven approaches that enhance precision, efficiency, and consistency
260 throughout the radiation therapy pipeline.

261 Central applications include deep learning–based medical image segmentation, where architectures
262 such as U-Net enable fast and accurate delineation of tumors and organs at risk, substantially reducing
263 clinician workload. We also explored AI-guided dose prediction, which leverages historical treatment
264 data to generate patient-specific radiation plans with improved speed and personalization. Finally,
265 we discussed emerging methods for synthetic data generation as a means to overcome limited data
266 availability, mitigate privacy concerns, and improve model generalizability.

267 Together, these advances illustrate the potential of AI to support more scalable, data-driven, and
268 patient-centered radiation therapy.

269 **6.2 Future Outlook**

270 The future of artificial intelligence promises to fundamentally transform how radiation therapy is
271 planned, delivered, and monitored. Large language models (LLMs), for example, may serve as
272 intelligent clinical agents capable of integrating multimodal data including medical imaging, genomic
273 profiles, and real-time biomarkers to generate personalized treatment recommendations. Advances in
274 segmentation, powered by vision foundation models, will enable more precise delineation of subtle
275 anatomical structures and tumor margins. Beyond dose prediction, deep learning models may directly

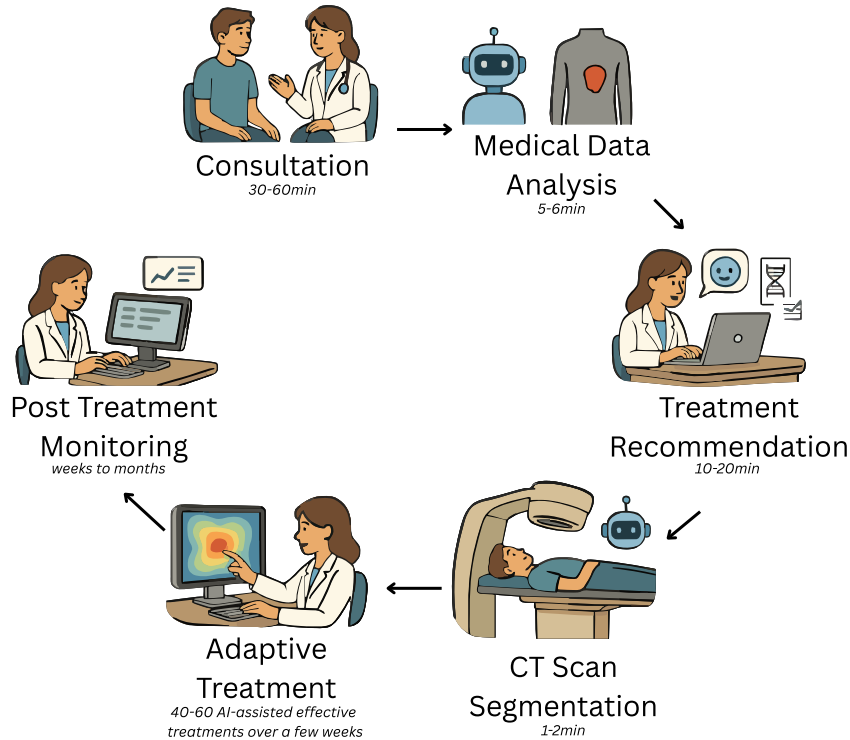


Figure 6: Future AI-Enhanced Radiation Therapy Workflow

infer machine-level treatment parameters, including beam orientations, fluence distributions, and multi-leaf collimator (MLC) settings, thereby streamlining the planning pipeline and minimizing manual intervention.

To envision this transformation, let us return to Mr. Harrison. In a future AI-augmented workflow as illustrated in Figure 6, his care pathway would be markedly accelerated and optimized. Following his initial consultation, an AI-based clinical agent could instantly synthesize his medical history and presenting symptoms to recommend a tailored treatment strategy. Advanced segmentation networks would automatically delineate his tumor and organs at risk with high precision, and dose prediction models would generate optimized radiation plans within minutes. During treatment, AI systems would monitor real-time anatomical changes, adapting the radiation plan dynamically to ensure precise targeting throughout each session. In post-treatment, automated tools would continue to track his response and flag any emerging side effects, enabling proactive care adjustments.

This shift from a slow and expert-driven workflow to a streamlined and AI-supported process represents more than a technical upgrade. It reimagines the entire patient journey by enabling faster, more consistent, and highly personalized care. As AI systems continue to evolve, they hold the potential to reduce disparities in access, ease the burden on clinicians, and deliver expert-level radiation therapy to patients like Mr. Harrison, regardless of where they receive treatment.

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